

MRI-Based Diagnosis and Classification of Alzheimer's Disease

Paper Overview

Topic: Early Alzheimer's Disease Diagnosis using Deep Learning on MRI Images

This review summarizes the paper 'A Deep Learning Framework for Early Diagnosis of Alzheimer's Disease on MRI Images'. The study presents an end-to-end MRI classification framework based on a custom CNN and a fine-tuned VGG16 model with preprocessing, augmentation, transfer learning, and automated diagnosis.

Paper Title	Author(s)/Year	Dataset/Source	Model/Technique	Key Findings
A Deep Learning Framework for Early Diagnosis of Alzheimer's Disease on MRI Images	Doaa Ahmed Arafa et al. 2024	Kaggle Alzheimer's MRI Dataset 6400 MRI images 5121 train / 1279 test	Custom CNN Fine-tuned VGG16 Transfer Learning 64×64 resizing Normalization Data Augmentation	CNN achieved 99.99% accuracy while VGG16 achieved 97.44%. Data augmentation improved class balance and reduced overfitting with relatively low computational complexity.

Summary

- Deep Learning Framework:** Combines preprocessing, augmentation, CNN and VGG16 for MRI-based Alzheimer's diagnosis.
- Image Preprocessing:** Resizes MRI images to 64×64 and normalizes pixel values.
- Data Augmentation:** Uses rotation, shifting, shearing, zooming and horizontal flipping.
- Model Performance:** Custom CNN outperformed fine-tuned VGG16.
- Limitations & Research Gap:** Binary-focused experiments, MRI-only modality, no Graph Neural Networks, limited explainability and cross-dataset validation.